

**REPORT FRONT SHEET**  
**MOLECULAR BIOLOGY, Master Course**  
**EXPERIMENTAL CELL BIOLOGY**

NAME OF EXPERIMENT:      Exercise 4 week 1 and 2: Fluorescent Imaging and Staining + Hygiene test

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DATE FOR PERFORMANCE OF EXPERIMENT:      15-06-2026 -- 26-06-2026

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TEACHER:      Ole Vang

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TEAM      Green

GROUP NUMBER:      3

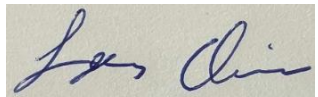
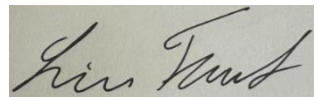
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Names and signatures

Andreas Giovanni Kristian  
Frilev Olesen,

Line Babette Trust,

Lars Eivind Bustamante  
Christoffersen



DATE(S) FOR HANDING IN THE REPORT TO THE TEACHER:

|                                |                  |                  |
|--------------------------------|------------------|------------------|
| Date for 1. Time<br>26.06.2026 | Date for 2. time | Date for 3. Time |
|--------------------------------|------------------|------------------|

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TEACHER COMMENTS:

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28/06/26: Minor correction should be done in 4A to get the report approved! Indicate the changes with red text.

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REPORT APPROVED BY TEACHER:

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Signature

Date

This is the front page for the reports and has to be attached to the report. The original front page should be attached when a revised report is handed in.

THE BIOLOGY STUDY BOARD

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## Introduction:

For studying and investigating cellular morphology and intracellular organization, fluorescence microscopy is used a technique where selective labeling specific cellular structures with fluorescent allows for visualization. In living cells, fluorescent dyes enable visualization while preserving cell viability, whereas fixation allows detailed examination of fixated cells. In the experiments, caco-r cells were exposed for 72 hours to resveratrol, the

While it is known that mammalian cells react to range of different chemical compounds from a wide variety of sources, such as the body of an organism or ingested food, the analysis of the effects of various compounds is complicated due to potential nonlinear combinatory effects.

In this exercise, we study the exposure of Caco-2 cells to Resveratrol, a widely study compound with a long range of purported effects and health benefits [citation: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9414442/>], and sodium butyrate (NaBu), a compound know for acting as a histone deacetylase inhibitor. 

## Methods:

For week 1, Caco-2 cells were grown in four separate Ibidi dishes and then subjected to a vehicle control of PBS and DMS, and then Resveratrol + PBS, NaBu + DMSO and finally Resveratrol + NaBu. The cells were then stained with Calcein Red-Orange, AM (Ex/Em: 577nm/590nm) to visualize live cells with intact plasma membranes, MitoTracker Green to visualize mitochondria (Ex/Em: 485nm/540nm) and Hoechst 33342 (Ex/Em: 360nm/460nm) to visualize the nucleus in live cells. Images were acquired on an Olympus FV-1200 confocal microscope using a 60x objective.

For week 2, Caco-2 cells were seeded on 8 separate cover slips coated with poly-D-lysine which were then treated with the same conditions as in week 1, using two discs for each treatment. The cells were then fixed with formalin and stained with Rhodamine-Phalloidin (Ex/Em: 540nm/565nm) to visualize the actin cytoskeleton and Hoechst 33342 (Ex/Em: 360nm/460nm) to visualize the nucleus. Images were acquired on an Olympus FV-1200 confocal microscope using a 60x objective.

Phase contrast is also used to check the viability of the cells for confocal microsscopy 

# Results

## Phase Contrast

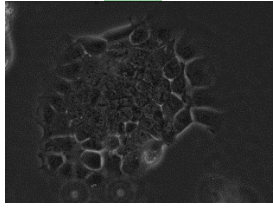
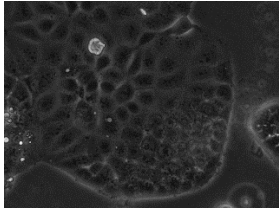
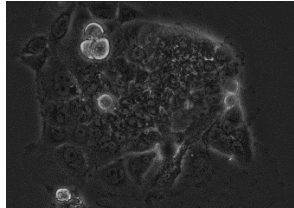
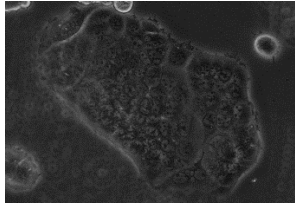
| DMSO NaB                                                                          | DMSO PBS                                                                          | Resv NaBu                                                                          | Resv PBS                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |

Table 1: Phase-contrast of the cell condition

Phase-contrast to check viability shows that there are a viable amount of cells to work with and all haven't all apoptosed

Live cell fluorescent imaging of nuclei

Fluorescence microscopy was used to examine Caco-2 cells following 72 h treatment with vehicle control, 20  $\mu$ M resveratrol, 1 mM sodium butyrate (NaBu), or the combination of both compounds (Figure 1). Hoechst staining clearly identified cell nuclei, while Calcein Red-Orange and MitoTracker Green allowed visualization of the cytoplasm and mitochondria.

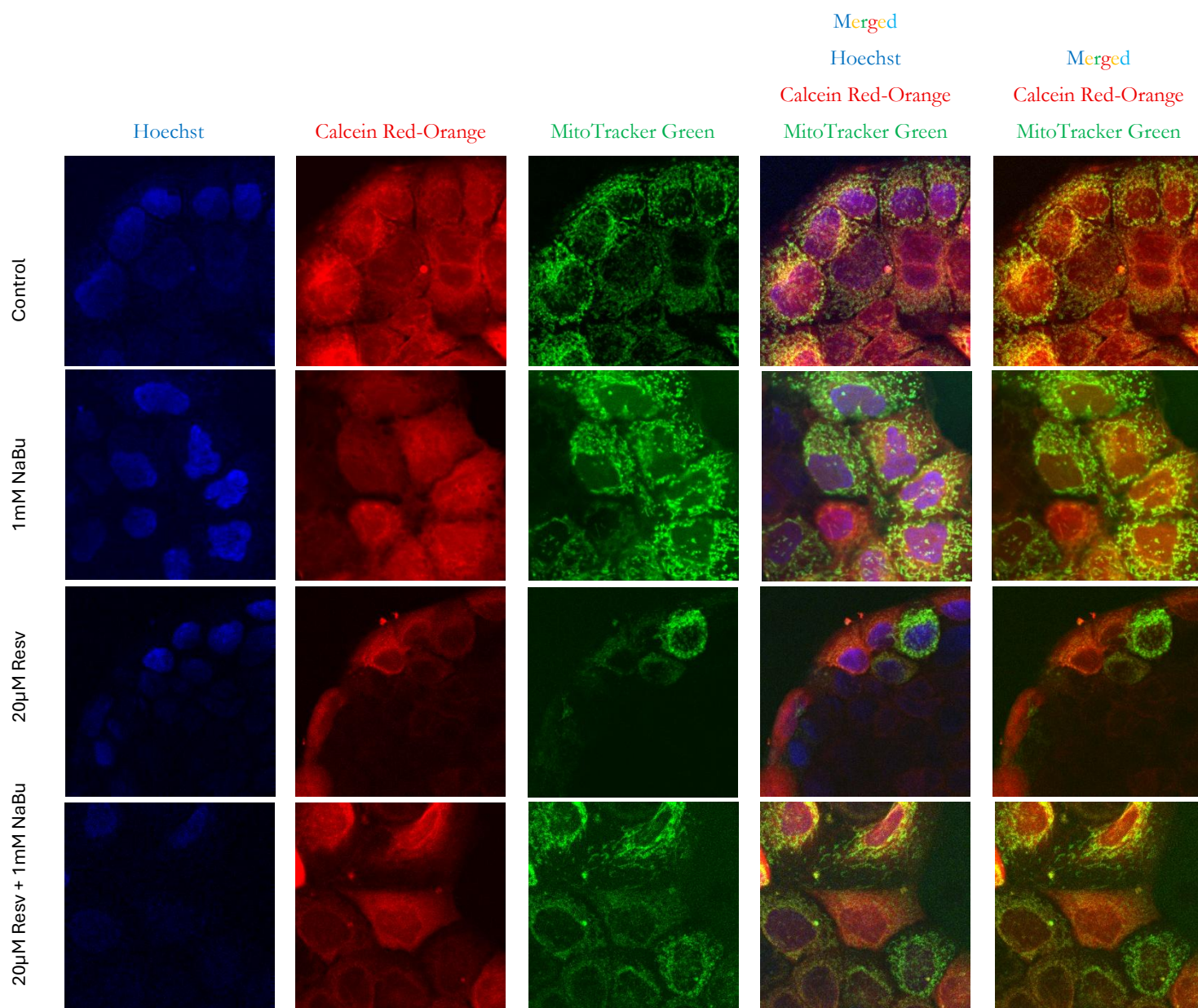
Fluorescence images were successfully acquired for the control, NaBu-treated and NaBu + resveratrol-treated cells. In these samples, the fluorescent staining was sufficiently strong to distinguish individual nuclei, cell morphology and the intracellular distribution of mitochondria. Overall, the cells remained adherent and displayed intact morphology, indicating that the majority of cells were viable at the time of imaging.

Unfortunately, the resveratrol-treated sample did not produce a detectable fluorescent signal because of a technical error during sample preparation or image acquisition. Consequently, representative fluorescence images could not be obtained for this treatment, preventing direct comparison between all four experimental groups, however fluorescence images for resveratrol-treated samples were borrowed from group blue, since hey also worked with the caco-2 cell line for this particular experiment.

Comparison of the treatment groups did not reveal obvious differences in nuclear morphology or mitochondrial distribution. Likewise, the overall cell morphology appeared comparable between the control, NaBu-treated and combined treatment groups under the imaging conditions used. Any subtle treatment-induced differences would require quantitative image analysis to be evaluated.

Images was captured with confocal microscope (Olympus FV-1200)

Fluorescence filters (maximas): Hoechst 33342: Excitation: 361 nm; Emission: 497 nm. Calcein RedOrange: Excitation: 577 nm; Emission: 590 nm Mito Tracker Green: Excitation: 490nm; Emission: 516 nm.



**Figure 1. Immunofluorescence analysis of Caco-2 cells.** Nuclei were stained with Hoechst. Calcein Red-Orange was used for visualization of the cytoplasm. MitoTracker Green to stain live intracellular mitochondria. Merged panels show overlapping channels. Images were captured with confocal microscope (Olympus FV-1200). Fluorescence filters (maximas): Hoechst 33342: Excitation: 361 nm; Emission: 497 nm. Calcein Red-Orange: Excitation: 577 nm; Emission: 590 nm Mito Tracker Green: Excitation: 490nm; Emission: 516 nm. Images were acquired using a 60× objective

## Quantification of nuclei in lice cells week 1

Nuclear area was quantified using ImageJ, from live-cell images acquired after Hoechst 33342 staining. Because a calibrated image scale was unavailable at moodle during analysis, the absolute nuclear areas reported by ImageJ should not be interpreted as true physical measurements. However, as all images were acquired at the same magnification and analyzed using identical settings, the measurements provide a valid basis for relative comparison between treatment groups.

Figure 3 display the mean nuclear area, which was similar across the three analyzed groups, with mean values of 293 for the control, 294 for the NaBu-treated cells, and 281 for the combined NaBu + resveratrol treatment. Although the NaBu-treated group displayed the highest mean value, the differences between the groups were small. The combined treatment showed the greatest variation in nuclear area, as reflected by the larger standard deviation and wider distribution of individual measurements. Overall, the distributions of nuclear area overlapped considerably between the three treatment groups, and no clear treatment-dependent changes in nuclear size were observed.

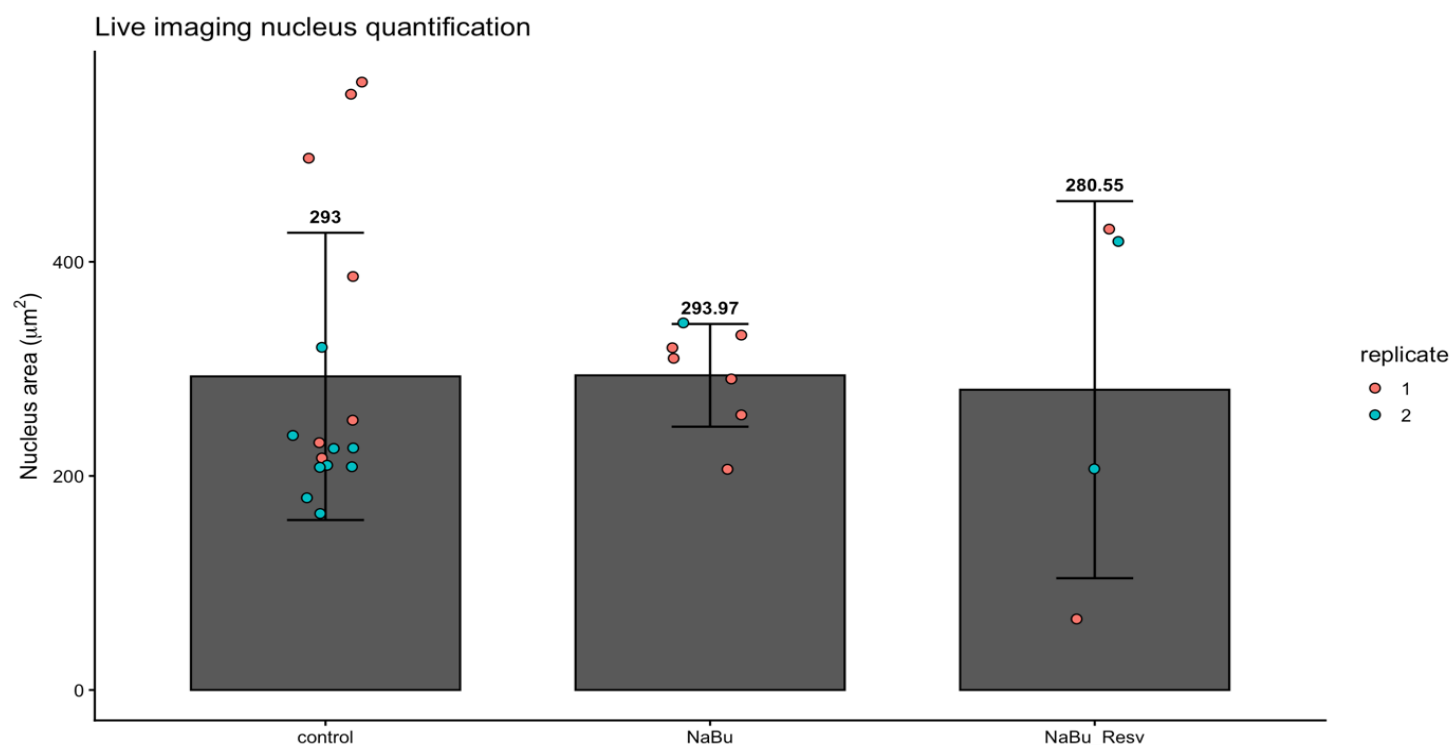


Figure 2. Quantification of nuclear area in live Caco-2 cells following treatment with sodium butyrate and resveratrol. Caco-2 cells were treated for 72 h with vehicle control, 1 mM sodium butyrate (NaBu), or the combination of 1 mM NaBu and 20  $\mu\text{M}$  resveratrol (NaBu + Resv). Live cells were stained with Hoechst 33342, and nuclear area was quantified from fluorescence images using ImageJ. Individual data points represent measurements from two biological replicates. Bars indicate the mean nuclear area pooled across replicates, and error bars represent the standard deviation.





## Fluorescence imaging of actin cytoskeleton



Following 72 h of treatment, Caco-2 cells were fixed and stained with Hoechst 33342 and rhodamine-conjugated phalloidin to visualize cell nuclei and polymerized actin, immunofluorescence images as displayed on figure 2.

Hoechst staining produced strong nuclear fluorescence in all treatment groups, allowing clear visualization of individual nuclei and confirming successful fixation and nuclear staining, allowing for further quantification of nuclei, further down in the report.

In contrast, the rhodamine-phalloidin staining produced only weak and diffuse red fluorescence with little discernible filamentous actin structure. Consequently, the actin cytoskeleton could not be reliably visualized in any of the treatment groups, preventing a qualitative comparison of actin organization between control, resveratrol-, NaBu-, and combination-treated cells.

Because the phalloidin staining was unsuccessful, no conclusions regarding treatment-induced changes in cytoskeletal organization could be drawn from this experiment. The successful Hoechst staining nevertheless confirmed the presence of intact nuclei in all samples and verified that the samples remained on the coverslips throughout the staining procedure.

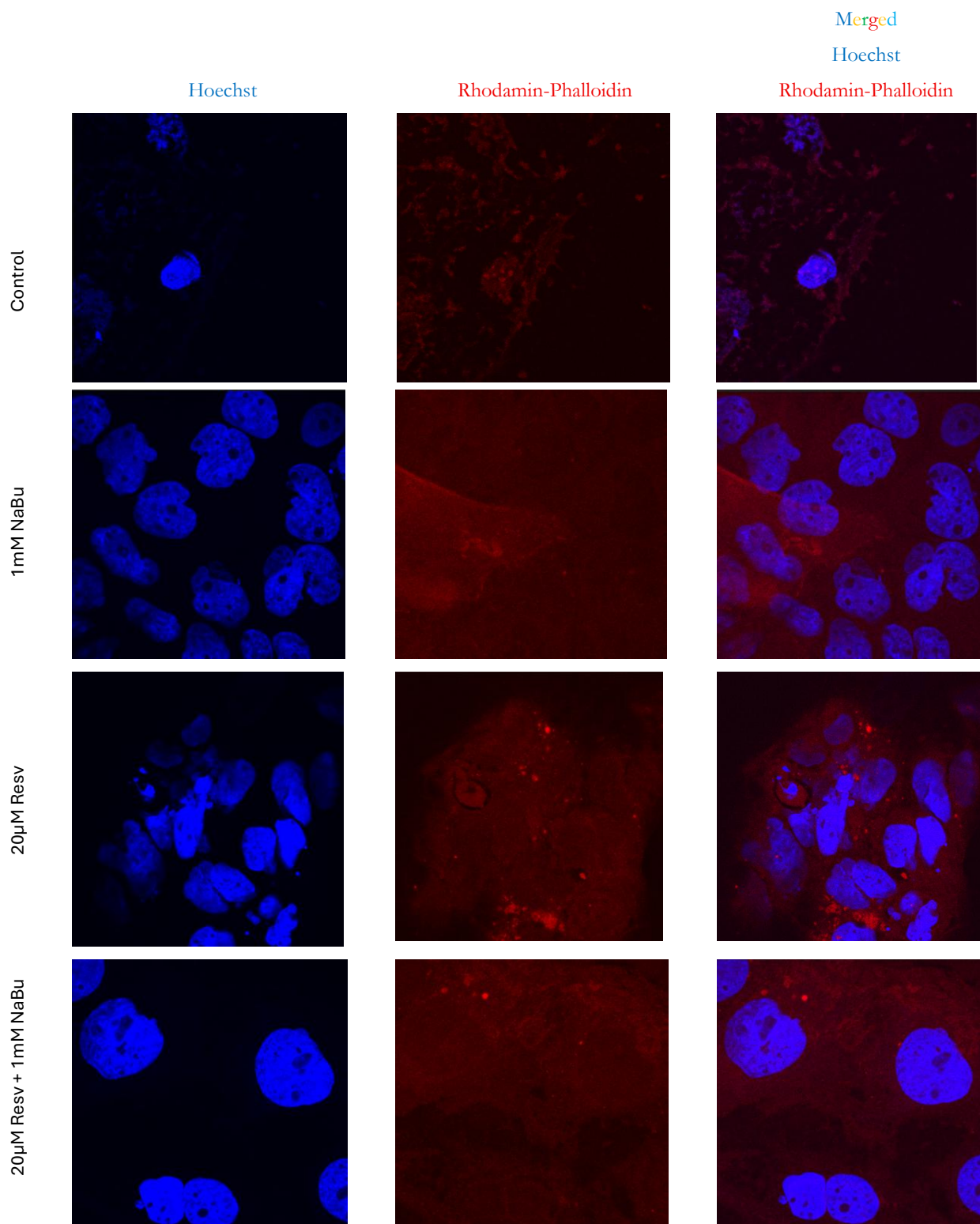


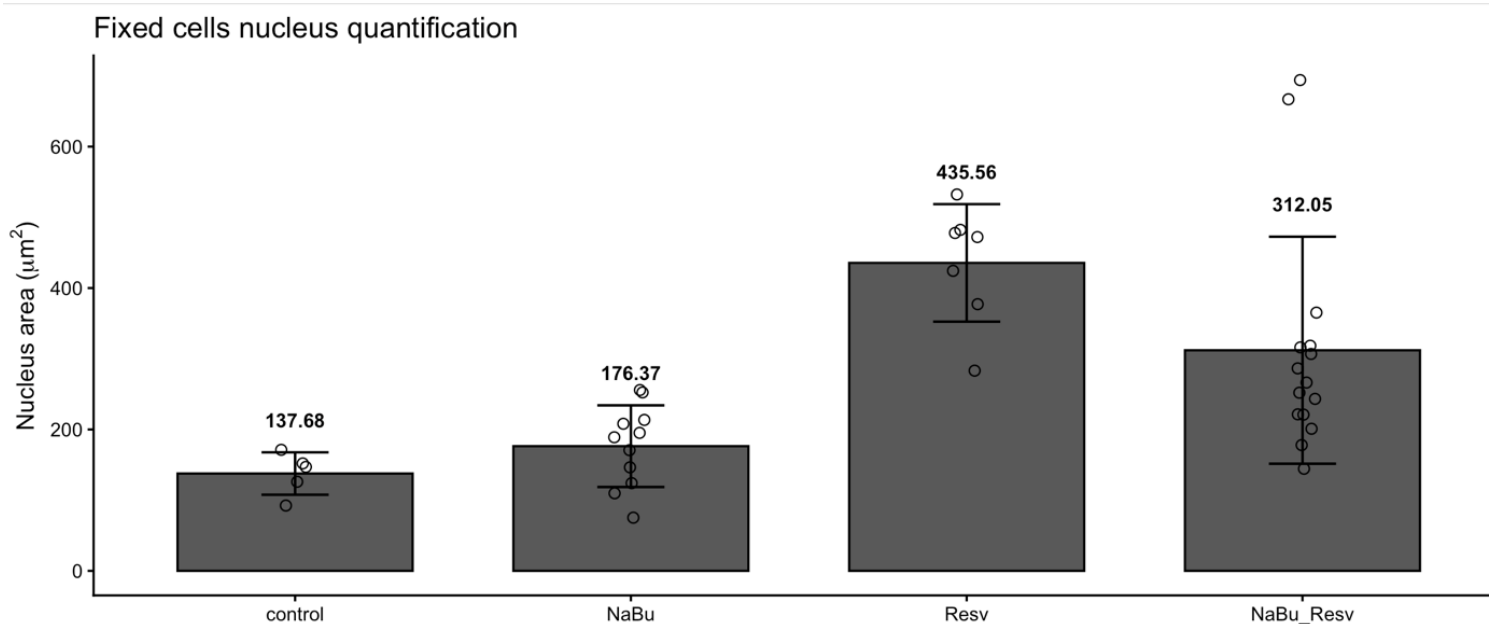
Figure 3. Fluorescence imaging of fixed Caco-2 cells stained for filamentous actin. Caco-2 cells were cultured on poly-D-lysine-coated coverslips and treated for 72 h with vehicle control, 20  $\mu$ M resveratrol, 1 mM sodium butyrate (NaBu), or the combination of resveratrol and NaBu. Cells were fixed in 10% formalin and stained with Hoechst 33342 to visualize nuclei (blue) and rhodamine-conjugated phalloidin to label filamentous (F-) actin (red). Merged images show the overlay of both fluorescence channels. Images were captured with confocal microscope (Olympus FV-1200). Images were acquired using a 60 $\times$  objective.



## Quantification of nuclei in fixed cells week 2

Nuclear area was quantified from fluorescence images of fixed Caco-2 cells stained with Hoechst 33342 as displayed in figure 4. In contrast to the live-cell analysis, greater differences in mean nuclear area were observed between the four treatment groups. The control group exhibited the smallest mean nuclear area ( $137.68 \mu\text{m}^2$ ), followed by the NaBu-treated group ( $176.37 \mu\text{m}^2$ ). Cells treated with resveratrol displayed the largest mean nuclear area ( $435.56 \mu\text{m}^2$ ), while the combined NaBu + resveratrol treatment resulted in an intermediate mean nuclear area ( $312.05 \mu\text{m}^2$ ).

The variability in nuclear area also differed between treatments. The NaBu + resveratrol group exhibited the widest distribution of measurements and the largest standard deviation, indicating considerable variation in nuclear size among individual cells. Although the resveratrol-treated group showed the largest mean nuclear area, the distributions of measurements partially overlapped across the treatment groups, particularly between the resveratrol and NaBu + resveratrol conditions.



**Figure 4. Quantification of nuclear area in fixed Caco-2 cells following treatment with sodium butyrate and resveratrol.** Caco-2 cells were treated for 72 h with vehicle control, 1 mM sodium butyrate (NaBu), 20  $\mu\text{M}$  resveratrol (Resv), or the combination of NaBu and resveratrol. Following fixation and Hoechst 33342 staining, nuclear area was quantified using ImageJ. Individual circles represent measurements of single nuclei. Bars indicate the mean nuclear area, and error bars represent the standard deviation.



## Discussion

Week 1 nucleus quantification showed relatively similar mean nuclear areas across the 4 different treatments. The combined treatment with Nabu + Resveratrol displayed greater standard deviation in nuclear size, than the other groups. A limitation of this analysis was the lack of image calibration, meaning that only relative rather than absolute nuclear areas could be compared. Furthermore, the resveratrol treatment could not be analyzed because of unsuccessful fluorescence staining, limiting the overall comparison between treatments.

Week 2 nucleus quantification revealed more pronounced differences between the four conditions, in the nuclear areas. Cells treated with resveratrol displayed the largest nuclear in mean, while the resveratrol in combination with NaBu exhibited largest standard deviation.

This could possibly be explained by cell cycle arrest that can be caused by resveratrol leading to granulation and increase in nucleus size. The resveratrol conditions also appear to have a larger variety of nucleus sizes which could be explained by the cell being at different stages of apoptosis. Largest variety being in the NaBu\_Resv condition which mirrors the Week 1 nucleus quantification.

Cell size comparison could also give an insight into the effect of resveratrol, however, issues arose when staining actin in the cell with rhodamine. Rhodamin not staining the actin filament made cell structure and spread difficult to impossible to identify



## Conclusion

The week 1 fluorescence gave us results with other than variation of nucleus sizes between the different conditions, the overall mean seemed to be relatively equal. Expectation were to have a larger variation in the mean so possibly either a mistake was made while creating the conditions for the first week or a possible change in concentration of resveratrol for future experiments.

The week 2 fluorescence gave us strong correlations between nucleus size in the cell and resveratrol, possible caused by cell cycle arrest. A suggestion could be the use of flow cytometry to analyze if the cells are halting at certain stages of the cell cycle. In regards actin staining, it is possible that the rhodamine used was expired. Other groups also reported failed staining of actin for week 2.



## Hygiene exercise (4b):

### Introduction

Hygiene is vital for proper and replicable cell biologic experiments. In these experiments, the media that we use to grow DLD-1 and Caco-2 cells contains Penicillin and Streptomycin, which helps keep bacteria out of the cultures. However, certain strains of bacteria, including resistant bacteria, and eukaryotic fungi might not care about these antibiotics and therefore it is important to be aware of proper hygiene when handling the cells and their flasks, especially as we handle them barehanded in this course.

In this experiment, each member of the team inoculates 8 LB-plates, according to 8 different levels of handwashing/sanitizing and the plates are then visualized and counted to give an overview of the transmittance of bacteria from human fingers to objects.



### Methods

LB plates were inoculated with fingertips according to each of the steps described in the lab protocol. There was a deviation in the inoculation where Lars used a more gentle touch and Line and Andreas used a more forceful touch.



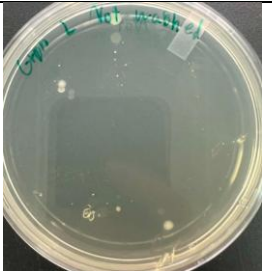
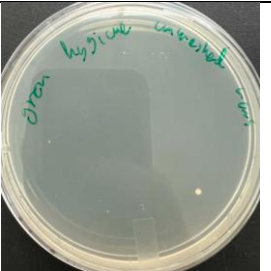
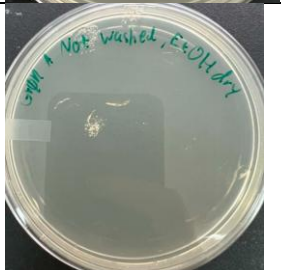
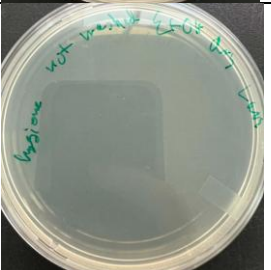
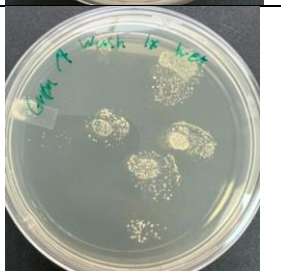
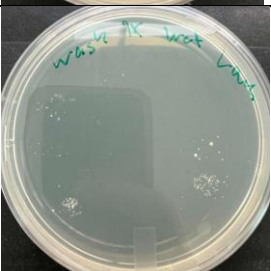
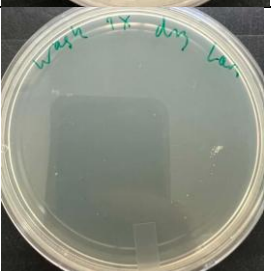
### Results

The results are visualized in table 1 and figure 1.

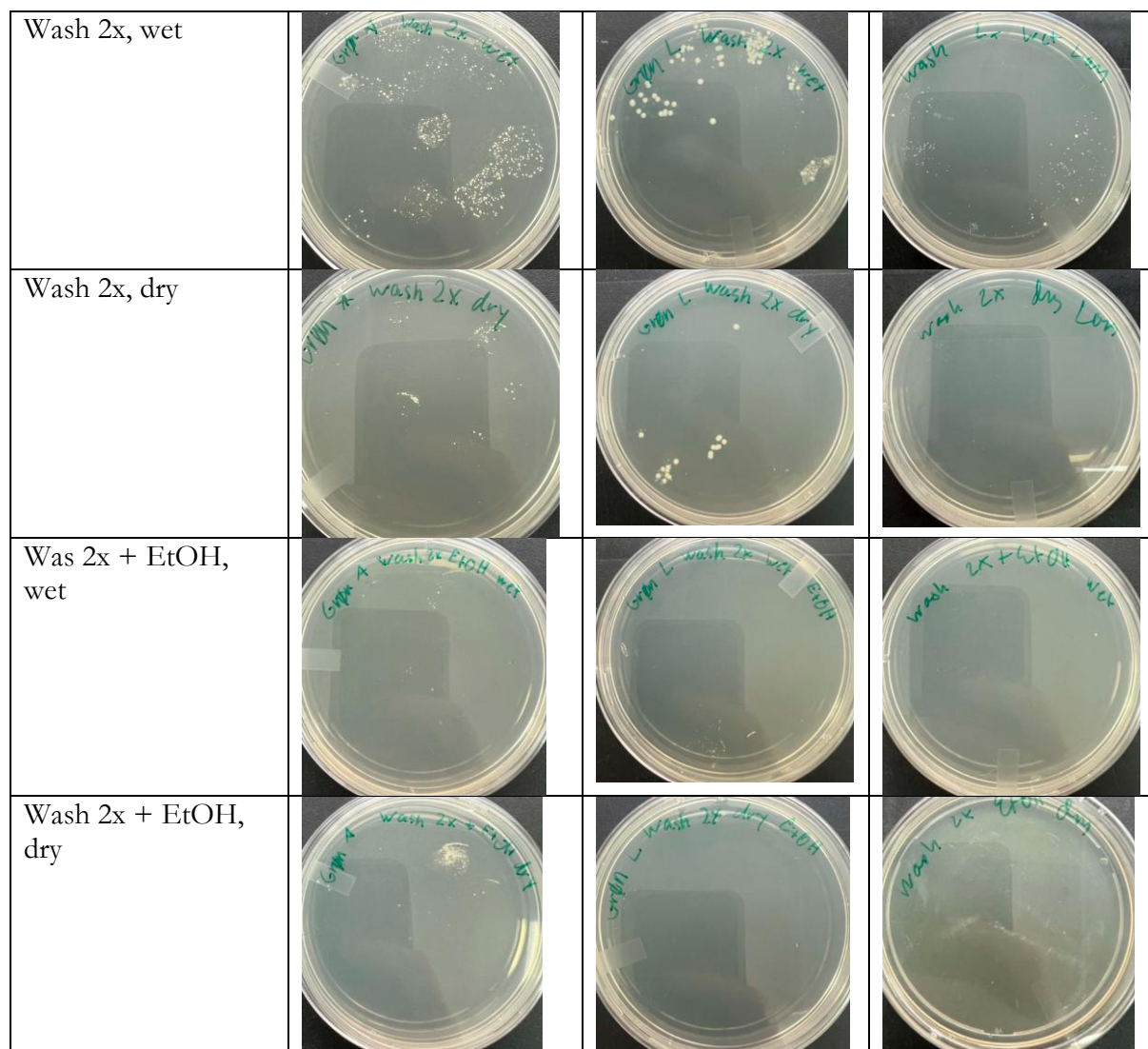
We see that for unwashed hands each subject inoculated bacteria onto the plate, while sterilizing the hands in ethanol and then drying significantly abolishes the amount of colonies. Despite previous sterilization with ethanol, washing the hands and then touching the agar with wet fingertips inoculates the plates with a very higher number of colonies. Subsequently touching the plates with dry fingers then reduces the amounts of colonies formed. This phenomenon is repeated during the second wash, where wet hands lead to a large increase in colonies and drying them reduces the amount of colonies. This is also seen at a less pronounced level for the final ethanol sterilization step.

|                              | <i>Andreas</i> | <i>Lars</i> | <i>Line</i> |
|------------------------------|----------------|-------------|-------------|
| <i>Not washed</i>            | Many small     | 14          | 26          |
| <i>Not washed, EtOH, dry</i> | Many small     | 1           | 0           |
| <i>Wash 1x wet</i>           | Many small     | Many small  | 31          |
| <i>Wash 1x, dry</i>          | Many small     | 18          | 14          |
| <i>Wash 2x, wet</i>          | Many small     | Many small  | 69          |
| <i>Wash 2x, dry</i>          | Many small     | 1           | 16          |
| <i>Was 2x + EtOH, wet</i>    | 12             | 1           | 30          |
| <i>Wash 2x + EtOH, dry</i>   | Many small     | 2           | 3           |

*Table 1. colony count on Lysogeny Broth-plates from Andreas, Lars and Line under different circumstances.*

|                       | Andreas                                                                             | Line                                                                                 | Lars                                                                                  |
|-----------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Not washed            |   |   |   |
| Not washed, EtOH, dry |  |  |  |
| Wash 1x wet           |  |  |  |
| Wash 1x, dry          |  |  |  |





**Figure 1.** *Lysogeny Broth-plates without antibiotics after 24 hours incubation for 37°C under different circumstances as indicated in column 1.*

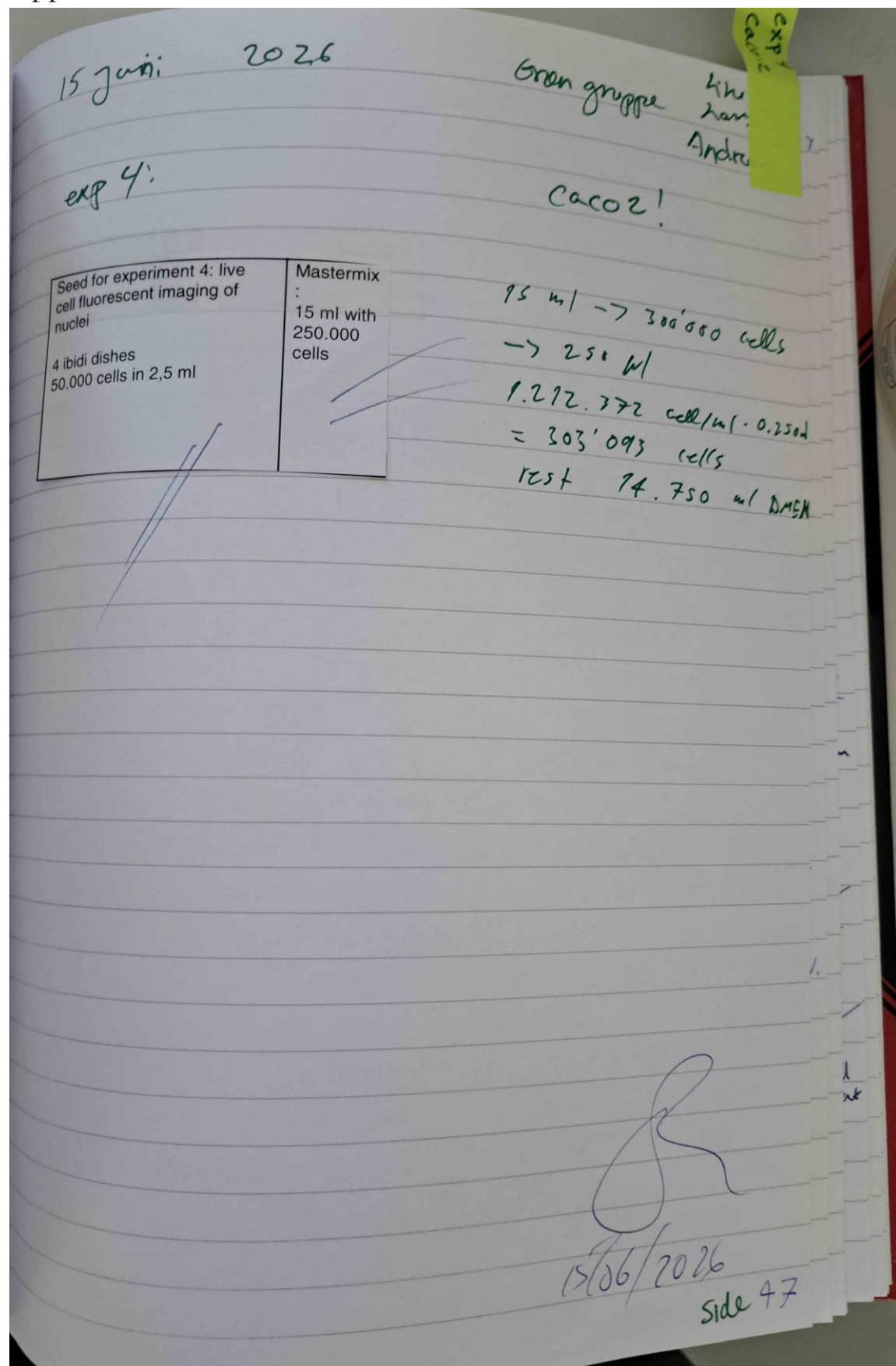


## Discussion

While the results from table 1 and figure 1 show that handwashing and ethanol sterilization has an effect on the number of bacteria transmitted from human fingers to objects, touching the plates with bare hands significantly increases the amount of bacteria transmitted, even if previous sanitation procedures had taken place. This means that even if hands are washed or sterilized in ethanol, care must be shown to make sure that hands are dry when handling sterile objects or cultures to avoid contaminating them with bacteria.



# Appendix: Lab book



16.06.2026 green team: Line, Lars, Andreas  
CaCo-2  
Experiment 4: Fluorescent imaging and staining

Stocks of:  
20  $\mu$ M resv + 1 mM PBS,  
~~20  $\mu$ M DMSO~~ + 1 mM PBS,  
20  $\mu$ M resv + 1 mM NaBu,  
vehicle DMSO + 1 mM NaBu

Volumes for stocks:  
1 mM NaBu: 5 ml medium + 5  $\mu$ L 1 M NaBu  
1 mM PBS: 5 ml medium + 5  $\mu$ L 1 M PBS  
20  $\mu$ M resveratrol: 5 ml medium + 1  $\mu$ L 100  
nM resv  
vehicle DMSO: 5 ml medium + 1  $\mu$ L pure  
MSO

10  $\mu$ L Res 100 nM  
40 ~~100~~  $\mu$ L DMSO

+ DMSO (1  $\mu$ L)  
+ DMSO (7  $\mu$ L)  
+ PBS (5  $\mu$ L)  
+ PBS (5  $\mu$ L)

- 4 flasks with 5 mL medium each are filled
  - we add 5  $\mu$ L PBS to all except for the 1 mM NaBu stock
  - ~~4 flasks with 5 mL medium each are filled with 20  $\mu$ M resv + 1 mM PBS, 20  $\mu$ M resv + 1 mM NaBu, 20  $\mu$ M resv + 1 mM PBS, 20  $\mu$ M resv + 1 mM NaBu~~
  - we do a 5x dilution of 100 mM resv. to a 20 mM resv in DMSO
  - we drain the ibidi dishes for old medium and add <sup>2.5  $\mu$ L</sup> of the stocks onto the dishes, ~~pipetting up and down~~ to resuspend
  - the dishes are put back to incubation
- 16/06/2018

17.06.2026 green team: Line, Lars, Andreas  
CaCo-2  
Experiment 4:  
Fluorescent imaging and staining

- 10-12 slips incubated in NaOH at room temp
- After 2 hours wash with Milli Q
- Cover with 100 ug/ml poly D lysine in h<sub>2</sub>O, incubate 10 min at room temp
- Remove poly D lysine, add 70% ethanol

Hygiene test:

From protocol by Vang et al

1. Every person needs 8 LB plates w/o antibiotics.
2. Stamp each fingertip to one LB-plate and label \*Not washed\*
3. Put both index fingertips (pegefingertip) in ethanol, 70% for 30s and let air dry.  
Stamp the index fingertips onto one LB-plate and label \*Not washed, EtOH, dry\*
4. Wash hands. With wet fingers stamp another LB-plate and label \*Wash 1x, wet\*
5. Dry hands as much as possible with a paper towel and let air dry at least 1 min.  
Stamp another LB-plate and label \*Wash 1x, dry\*
6. Wash hands. With wet fingers stamp another LB-plate and label \*Wash 2x, wet\*
7. Dry hands as much as possible with a paper towel and let air dry. Stamp another LB-plate and label \*Wash 2x, dry\*
8. Wipe hands with ethanol and while hands are wet with ethanol stamp another LB-plate and label \*Wash 2x+EtOH, wet\*
9. Let hands air dry. Stamp another LB-plate and label \*Wash 2x+EtOH, dry\*
10. Incubate at 37°C O/N
11. Next day: Count and record colonies on each plate.  
Take pictures of each plate for documentation (use your camera phones).

both steps in FBV

→ outside LAF bench, incubated in petri dish, 2M NaOH

→ seal petri dish potentially,

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18.06.2026 green team: Line, Lars, Andreas

CaCo-2

Experiment 4:

Fluorescent imaging and staining

Seed cells on cover slips (for week 2)

1. Split CaCo-2 cells (after preparing slips)

Prepare slips:

- Place poly-D-lysine coated cover slips in 8 wells of 12 well plate. Ensure that upward facing side contains poly-D-lysine. Let remaining alcohol evaporate (placed to the side in well).

2. Split CaCo cells

- Use cells trypsinized in Exp 1 to seed. Seed in 8 wells of a 12-well plate, 50.000 cells per well in a total of 2 mL medium/well, i.e. 25.000 cells/ml

8 wells  $\rightarrow$  make 10\*2 ml for 500.000 cells.

50.000 per bundle

$\rightarrow$  cells are splitted and Coulter counter is used to find cell density

$$\begin{array}{r} 1154 \\ 1134 \\ \hline \text{average} \\ 1144 \end{array}$$

$$1144 \cdot 51 \cdot 2 = 116.688 \text{ cells per ml}$$

$$25000 \text{ cells per well} = \frac{25.000}{116.688} = 0,214 \text{ ml cell suspension}$$

$$2 \text{ ml media} - 0,214 \text{ ml cell suspension} = 1,786 \text{ ml}$$

$$\frac{300.000}{116.688} = 2,57 \text{ ml cell suspension}$$

$$\text{Mastermix} = \frac{500.000}{116.688} = 4,285 \text{ ml cell suspension}$$

5,715 ml media

= total volume of 10 ml

18/6/26

50

15.06-2026

We accidentally made a suspension of 70 ml with  
500'000 cells  $\rightarrow$  50'000 cells/ml instead of  
25'000 cells/ml, pipetted the x2 solution  
into A1 of 12 well plate, we shook it up  
again & poured more media in suspension for  
proper dilution.

19.06.2026 green team: Line, Lars, Andreas  
CaCo-2

#### Experiment 4:

Fluorescent imaging and staining:  
1. Microscope in room 28A.2-37: Leica DM  
IRB / EVOS M5000 or confocal Olympus  
microscope (28AB.0-07)

2. Look at cells in phase contrast  
microscopes to gain information on viability.  
3. If cells are viable and adhere well to  
bottom, then stain cytosol with Calcein Red-  
Orange, AM and mitochondria with  
MitoTracker Green

a. Remove media – wash 2x with fresh media  
(assuming this is 2.5 ml)

b. Add 2 mL medium / dish containing

i. Add 2  $\mu$ L 800  $\mu$ M Calcein Red-Orange, AM /  
2 mL media.

ii. Add 2  $\mu$ L 400  $\mu$ M Mito Tracker Green / 2 mL  
media.

4. Incubate for 60 min

5. After 45 min with Calcein and MitoTracker,  
counter stain with Hoechst 33342

a. Use 2 drops / 2 mL (1  $\mu$ L 10 mg/mL Hoechst  
33342 / 2 mL media)

6. Incubate the cells for additional 15 min.

7. Remove media and add PBSw with 10 mM  
glucose instead.

8. Then check cells under microscope to  
assess good staining.

9. Magnification: Use the 20x and 40x  
objectives.

10. Fluorescence filters:

a. DAPI filter (white light, blue filter) for  
Hoechst 33342 (Ex 360/40 nm / EM  
460/40 nm)

b. Rhodamine filter for Calcein RedOrange (Ex  
577/10 nm; Em: 590/20 nm)

c. GFP filter for Mito Tracker Green (Ex 485/20  
nm / EM 540/25 nm)

11. Fading. Be aware that especially red  
colors fade quickly. For our purposes,  
Hoechst is quite stable. Nevertheless, expose  
the areas of interest as short time as  
possible to fluorescent light, before acquiring  
images.

12. Identify the cells with phase contrast.  
Locate cells that do not touch neighboring  
cells. Identify the relevant focal plane. Switch  
to the fluorescence and acquire

Cell Obs:

10  $\mu$ L  
710  $\mu$ L  
710  $\mu$ L

image. For more extensive microscopy, it is  
also possible to acquire several  
pictures at the same position, but you might  
have to increase the exposure time.  
Be sure to acquire images of at least 20 nuclei  
of each condition.  
13. Label images with group, well and dye  
interrogated.

#### Fluorescence filters (maximas):

Hoechst 33342: Excitation: 361 nm;  
Emission: 497 nm.

Calcein RedOrange: Excitation: 577 nm;  
Emission: 590 nm

Mito Tracker Green: Excitation: 490nm;  
Emission: 516 nm

Treatment with NaBu and Resv:

2 ml volume for wells (2 wells for each  
treatment), ~~hidi~~ dishes used 2.5 ml, we need  
2 well x 2 ml/well  $\rightarrow$  4 ml, make 5 ml total.  
Use same concentrations for last week.

Make a new 5x resveratrol dilution in DMSO  
(100 mM  $\rightarrow$  20 mM)

Stocks:

Ctrl Vehicle PBS+ vehicle DMSO:

5 ml medium + 5  $\mu$ L 1 PBS + 5  $\mu$ L DMSO

1 mM NaBu + DMSO:

5 ml medium + 5  $\mu$ L 1 M NaBu + 5  $\mu$ L DMSO

PBS + 20  $\mu$ M Resv:

5 ml medium + 5  $\mu$ L PBS + 5  $\mu$ L 20 mM resv

1 mM NaBu + 20  $\mu$ M resveratrol:

5 ml medium + 5  $\mu$ L 1 M NaBu + 5  $\mu$ L 20 mM  
resv

Week 2

5x dilution

120  $\mu$ L DMSO

3  $\mu$ L Resv

→ 0 0 0  
→ 0 0 0  
→ 0 0 0  
→ 0 0 0

12  
well  
plate

DMEM

52

*[Signature]*



## Flourescene Image

DMSO + PBS: rød + med små blå prikker  
stykker af det blå bliver mindre ved  
forstørrelse.

30

A: PBS + DMSO 30 - B: lode 30

PBS + DMSO 31

NaBO + DMSO: 32

rød og klart blå kerne. Minder lidt  
om undersiden af en blodsporette

NaBu + DMSO: 33

PBS + resv Nothing found, Error might have  
occured.

NaBu + resv: 34 dish B: Svagt signal

NaBu + resv: 35 dish D:

22.06.2026 green team: Line, Lars, Andreas  
CaCo-2

Experiment 4:

Fluorescent imaging and staining:

1. Inspect the cells in the microscope. Use phase contrast 10x or 20x
2. Remove all medium and quickly add 0.5mL fixative (in fume hood, 10% formalin) and incubate for 20 min at room temperature.
3. Remove the fixative and wash three times with PBS for 5 min. Discard fixative appropriately.

Cover petri dishes with tinfoil from now on!

Stain with Rhodamin-Phalloidin:

- Mix in an Eppendorf tube 1  $\mu$ L Rhodamine-Phalloidin to 1 mL 1% BSA in PBS, add 100  $\mu$ L per cover slip. Incubate for 30 min at room temperature.
- If the solution is prepared in advance, seal the Eppendorf tube with tinfoil and keep it on ice.
- Wash the cover slips 2 x 5 min in PBS.
- Remove cover slips from the petri dish with scalpel and mount them on microscope slides with a drop of mounting media between the slide and cell side of the cover slips.

- Allow mounting media to cure overnight (minimum 12 hours) at room temperature. Seal the edge of the cover slip with nail polish, brand and color is optional. Let the samples dry in petri dishes covered with tinfoil and store them in the refrigerator.
- The samples may be kept in the refrigerator for several days. If there is a need for longer storage, this may be done at  $-20^{\circ}\text{C}$ .

→ whit with

with

poly-D side into drop of mounting

Observations:

dead cells floating in wells  
but also live adhering

22/06/2026 JL

89

23.06.2026 green team: Line, Lars, Andreas  
CaCo-2

Experiment 4:

Fluorescent imaging and staining:

Seal the edge of the cover slip with nail polish. Let the samples dry in petri dishes covered with tinfoil and store them in the refrigerator.

Fluorescence Microscopy with Ole1  
Magnification: Use the 40x and perhaps the 60x objectives (needs oil)

Fluorescence filters (Leica DM IRB):

a. DAPI filter: Excitation: BP 340-380;  
Emission: LP 425,

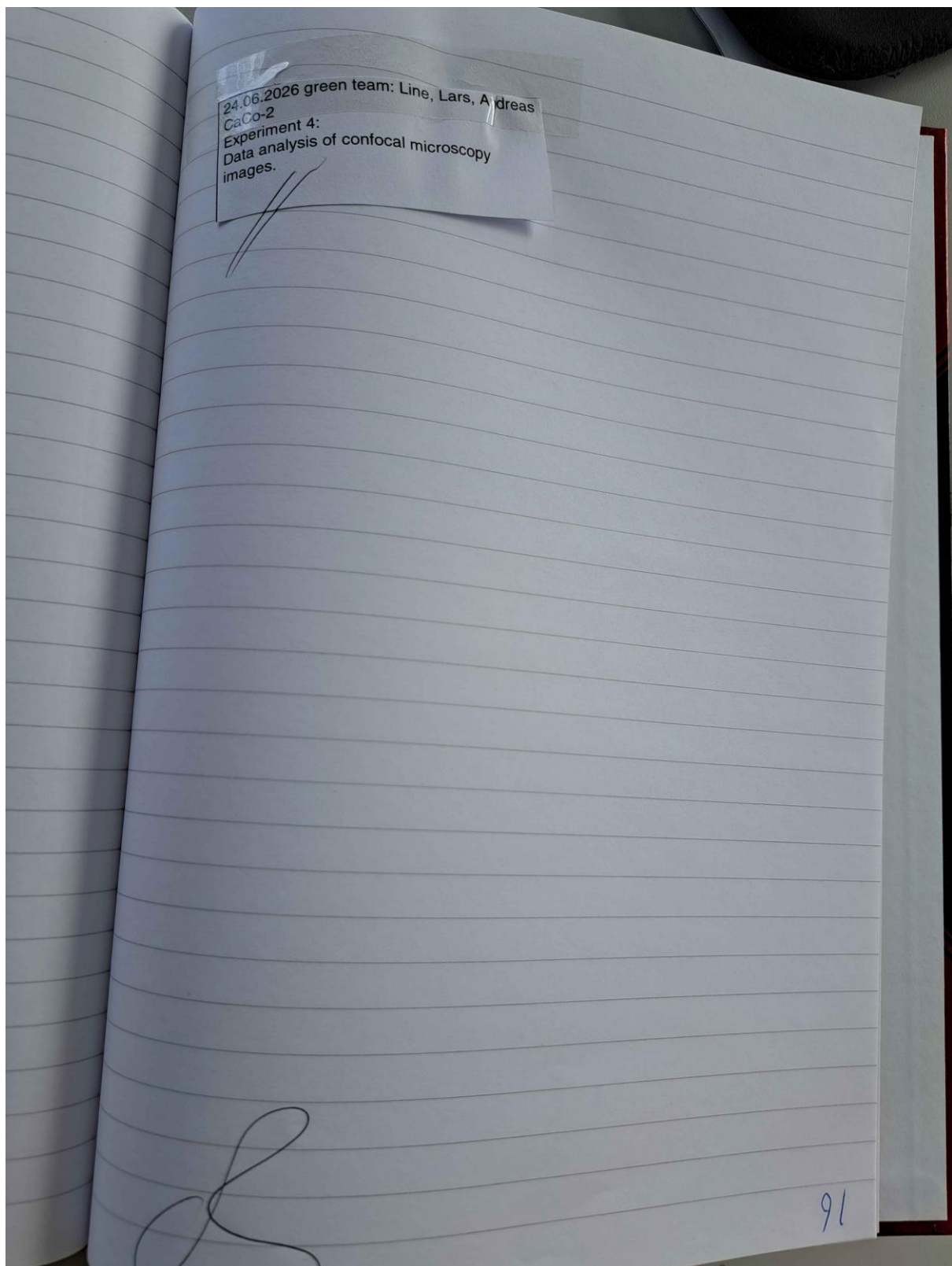
b. Rhodamine filter: Excitation: BP 515-560;  
Emission: LP 590

Fading. Be aware that especially rhodamine fades quickly. Therefore, expose the areas of interest as short time as possible to fluorescent light, before acquiring images.

Identify the cells with phase contrast.  
Identify cells which are placed individually.  
Find the relevant focus. Switch to the fluorescence and take pictures. It is possible to take several pictures at the same position, but you might have to increase the exposure time (start with 15 ms, aperture 1).

Acquire the image of the DAPI staining at the exact same position (i.e. do NOT move the slide!).





| image | Nucleus Area |
|-------|--------------|
| 1     | 386,255      |
| 1     | 496,671      |
| 1     | 567,988      |
| 1     | 556,732      |
| 1     | 230,955      |
| 1     | 252,195      |
| 1     | 216,598      |
| 2     | 237,82       |
| 2     | 209,977      |
| 2     | 226,233      |
| 2     | 208,043      |
| 2     | 208,461      |
| 2     | 320,288      |
| 2     | 179,572      |
| 2     | 164,709      |
| 2     | 225,536      |
| 3     | 206,108      |
| 3     | 257,126      |
| 3     | 319,852      |
| 3     | 290,58       |
| 3     | 331,404      |
| 3     | 309,956      |
| 4     | 342,765      |
| 5     | 430,46       |
| 5     | 66,194       |
| 6     | 206,666      |
| 6     | 418,89       |

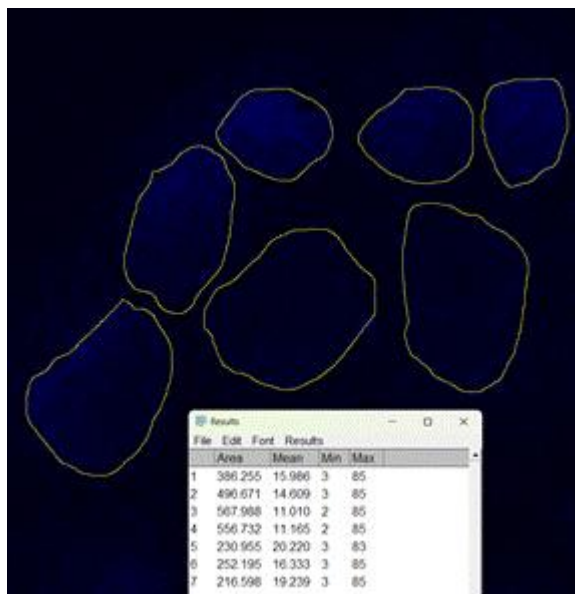
*Figure 1: The Nucleus Area's used to make comparisons for the confocal of week 1.*

Confocal notes: The magnification is x60, due to not having the measurement for x60 the number for the area is not correct however, this wont prevent us from comparing the sizes between the cells



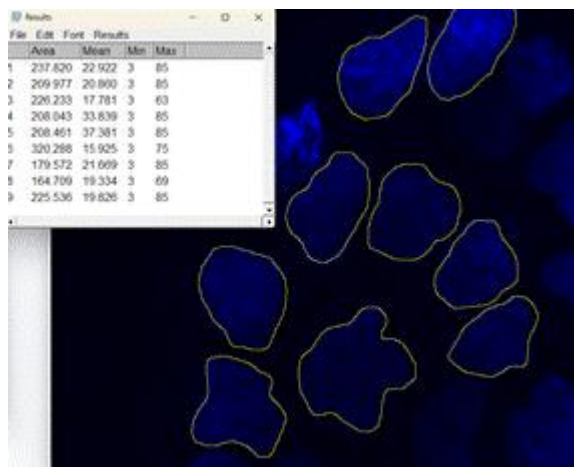
1 DMSO + PBS:

Nucleus:



2 PBS + DMSO

Nucleus



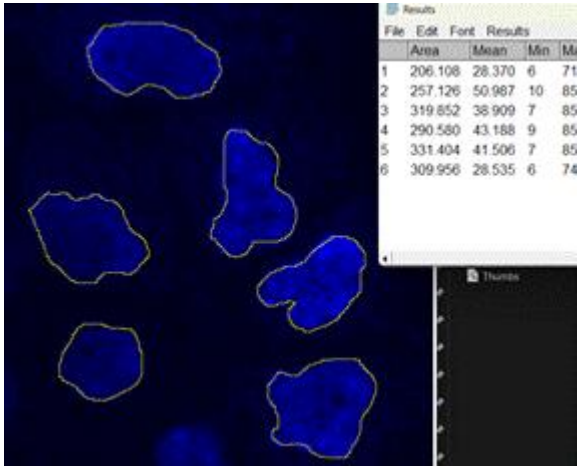
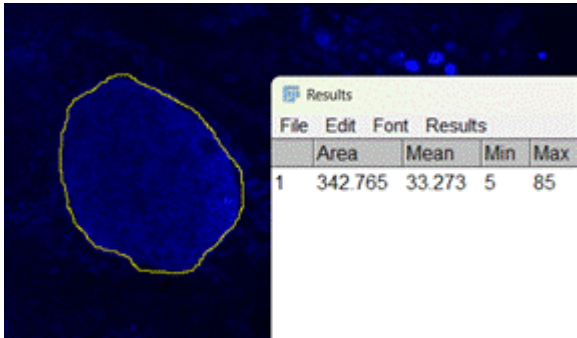
|                                                                                     |  |
|-------------------------------------------------------------------------------------|--|
|                                                                                     |  |
| 3 NaBu + DMSO<br>Nucleus                                                            |  |
|   |  |
| 4 NaBu + DMSO<br>Nucleus                                                            |  |
|  |  |
| 5 PBS + Resv<br>Nucleus                                                             |  |

Table 1: Table containing the raw data and images of the cells and how the area was gathered in imageJ.

Confocal Data week 2:

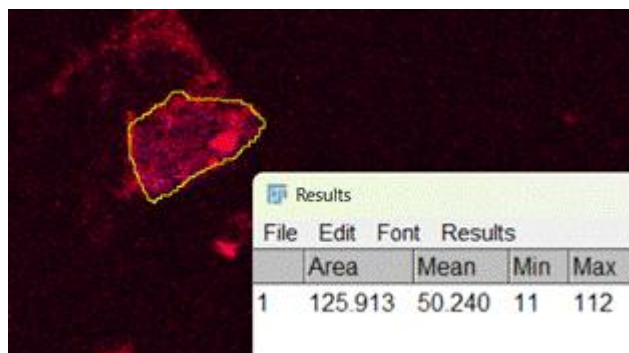
| Image | Nucleus Area |
|-------|--------------|
| 1     | 125,913      |
| 2     | 171,278      |
| 3     | 151,815      |
| 4     | 92,469       |
| 5     | 146,902      |
| 6     | 532,303      |
| 7     | 377,003      |
| 8     | 482,349      |
| 9     | 477,731      |
| 9     | 283,279      |
| 9     | 472,8        |
| 9     | 424,187      |
| 10    | 188,806      |
| 10    | 213,409      |
| 10    | 255,906      |
| 11    | 208,217      |
| 11    | 170,755      |
| 11    | 195,253      |
| 12    | 252,23       |
| 12    | 109,771      |
| 12    | 124,094      |
| 12    | 146,484      |
| 12    | 75,097       |
| 13    | 666,991      |
| 13    | 693,928      |
| 13    | 243,134      |
| 13    | 318,284      |
| 14    | 316,071      |
| 14    | 286,485      |
| 14    | 220,832      |
| 14    | 221,354      |
| 14    | 200,864      |
| 14    | 144,393      |
| 15    | 306,906      |
| 15    | 365,346      |
| 15    | 251,829      |
| 15    | 177,847      |
| 15    | 266,448      |

Figure 2: The Nucleus week 2 area data used to make comparisons. Repeat of the numbers on the left side are multiple nuclei measured on the same image.

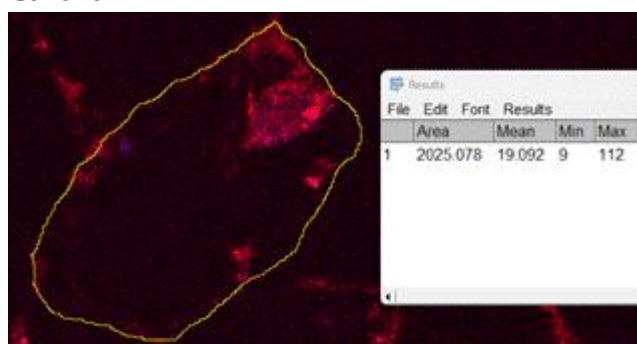
Confocal: Since we do not have a x60 scale, we will use a x20 and do a comparison of the sizes collected through image]

1. Control PBS + DMSO

Nucleus:

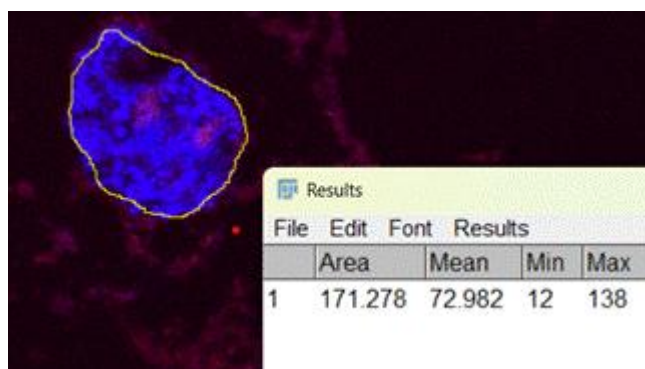


Cell size:

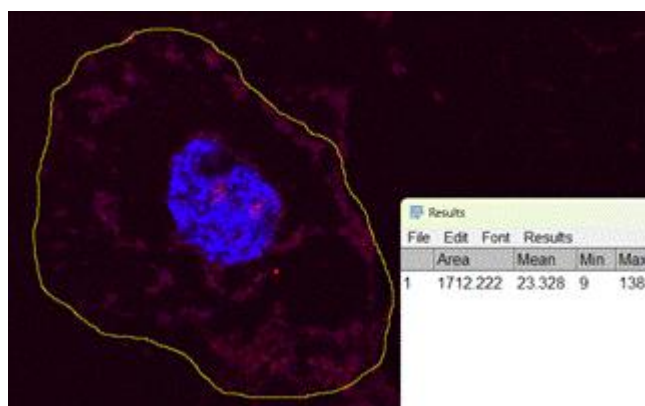


2. Control:

Nucleus:



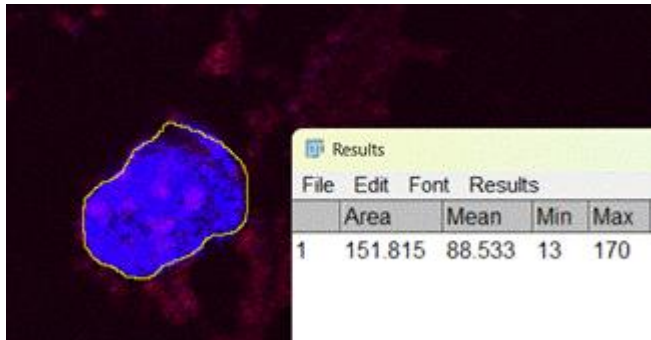
Cell size:



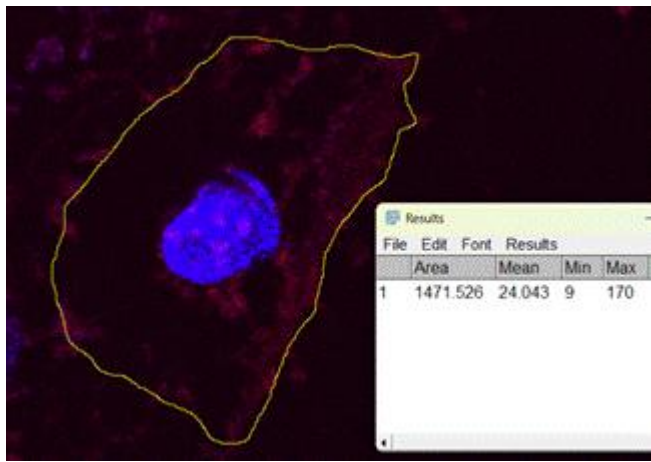


3. Control:

Nucleus:



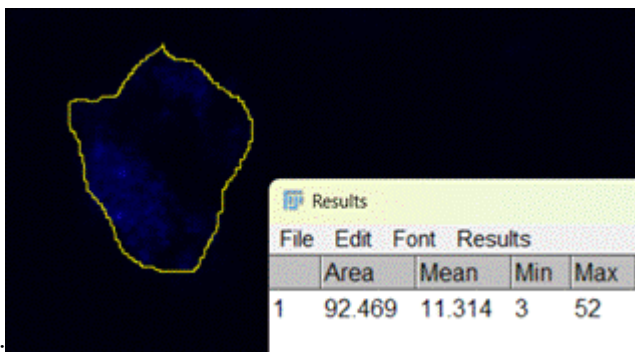
Cell size:

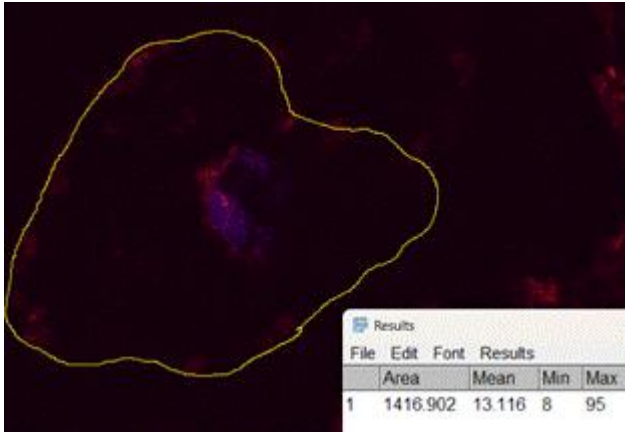


4. Control:

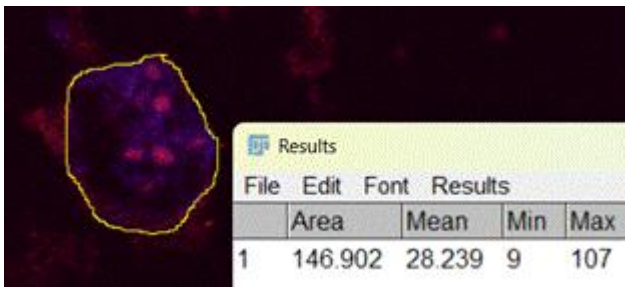
Nucleus:

Cell size:

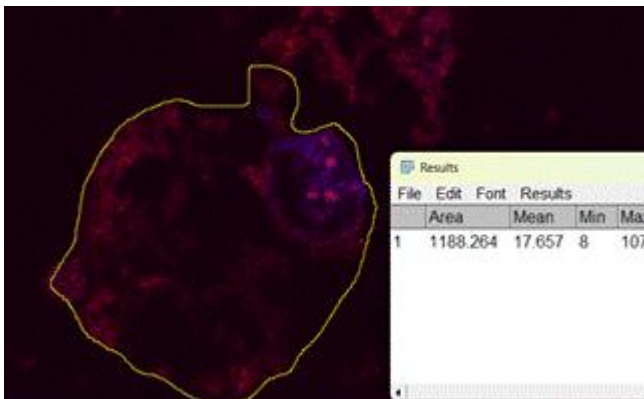




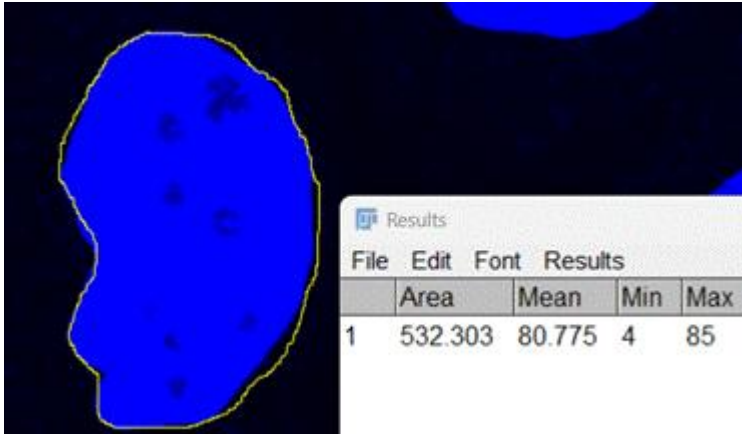
5. Control:  
Nucleus:



Cell size:



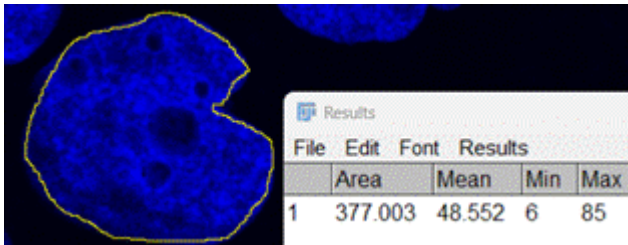
6. Resv+ PBS:  
Nucleus:



Cell size: Red oversaturated.

7. Resv+ '+ PBS:

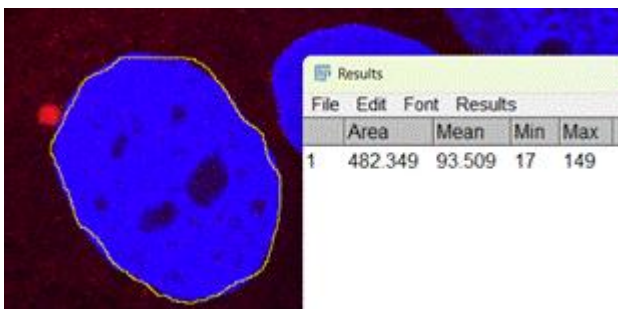
Nucleus:



Cell size: Not clear enough

8. Resv + PBS:

Nucleus:

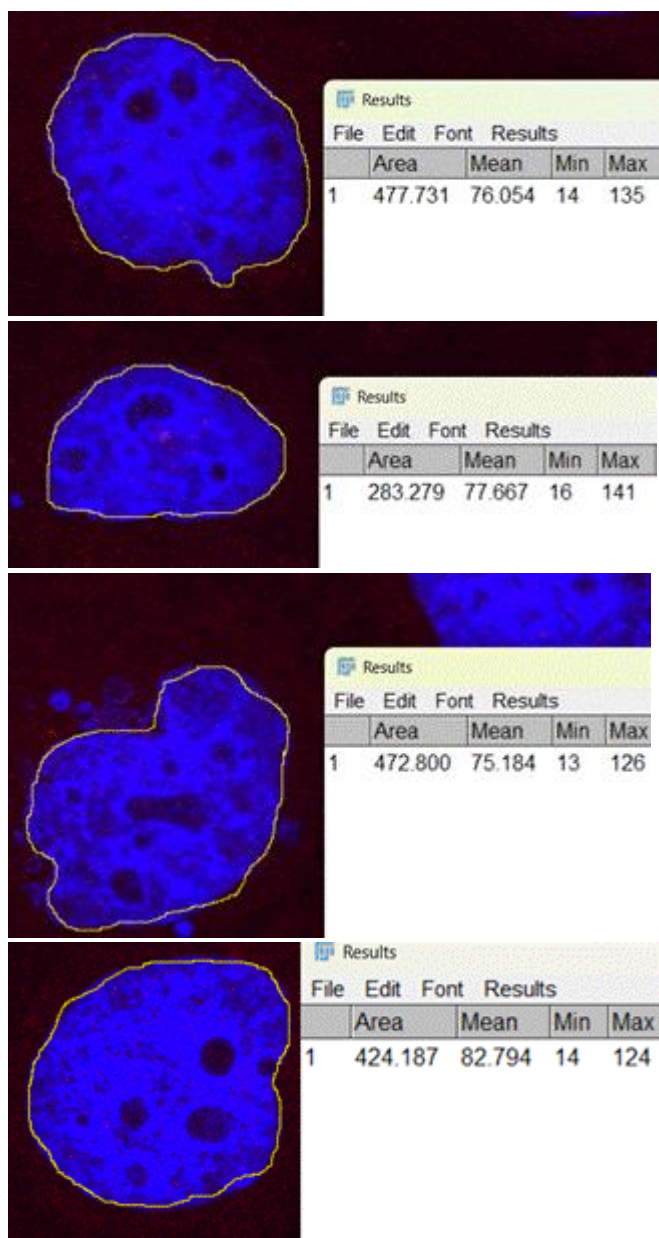


Cell size:

Unclear

9. Resv + PBS:

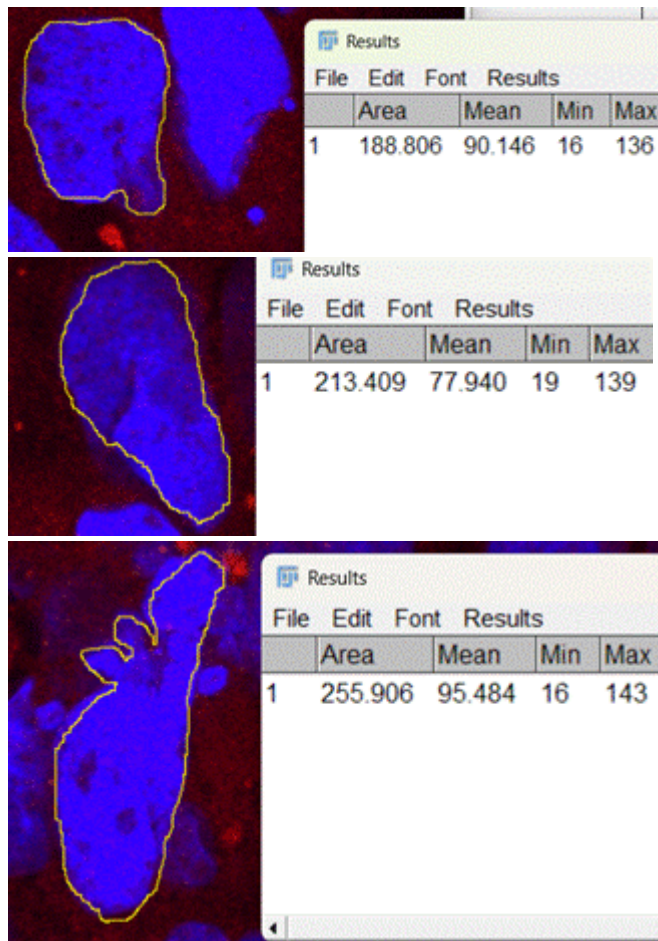
Nucleus:



Cell size: Not clear enough

10. NaBu + DMSO:

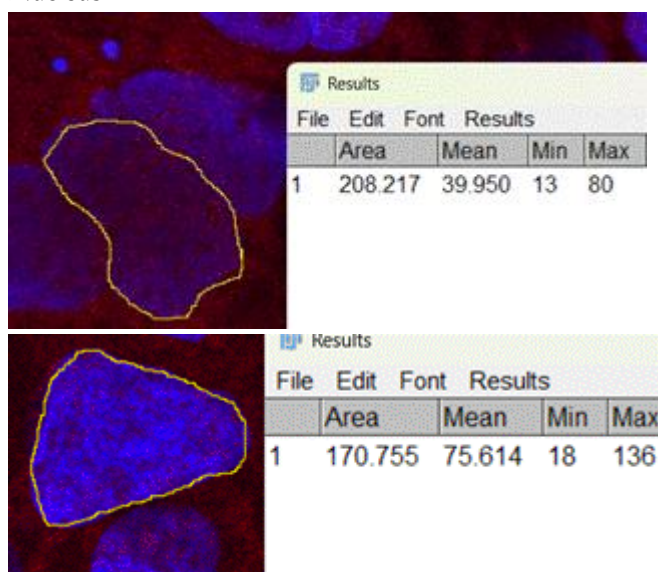
Nucleus:



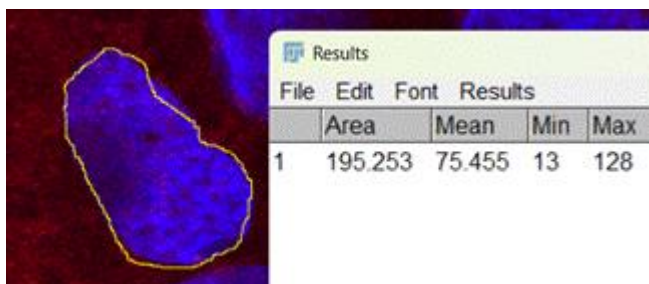
Cell size: Unclear

11. NaBu + DMSO:

Nucleus:

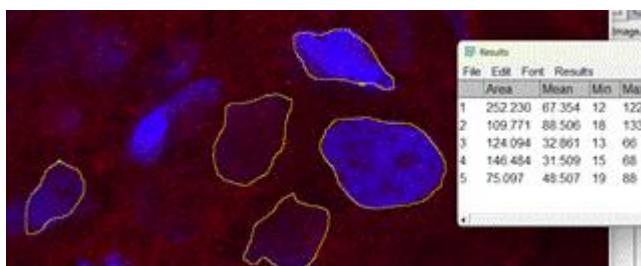






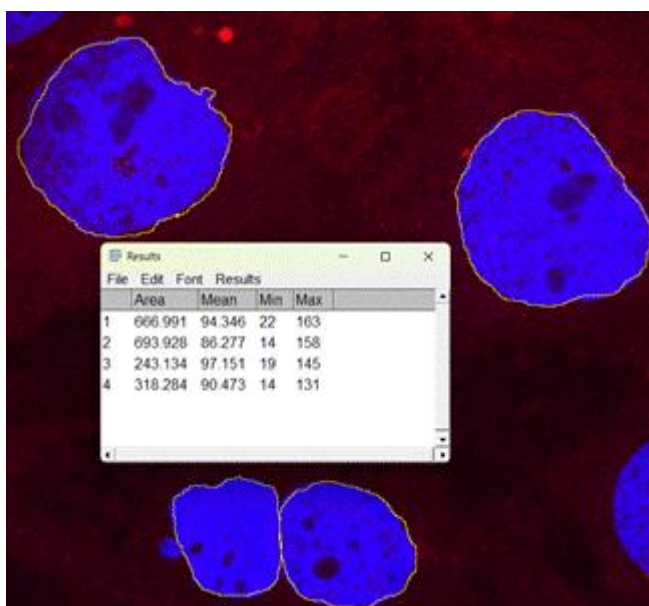
12. NaBu + DMSO:

Nucleus:

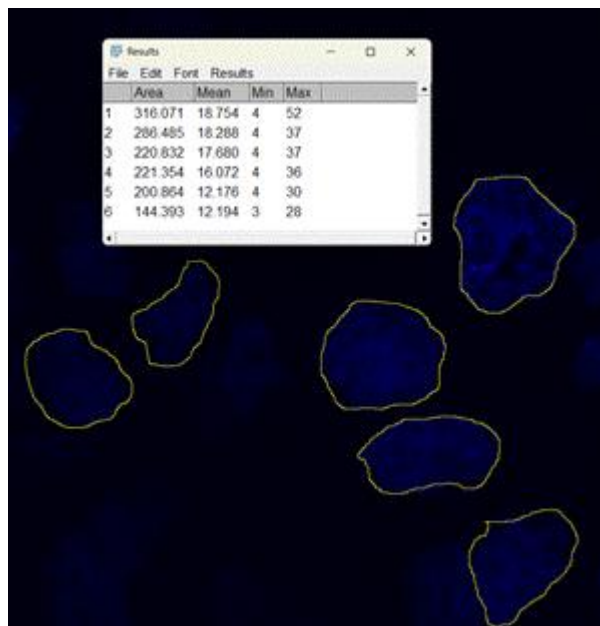


13. NaBu + Resv:

Nucleus:



14. Nabu + Resv:  
Nucleus:



15. NaBu + Resv:  
Nucleus:

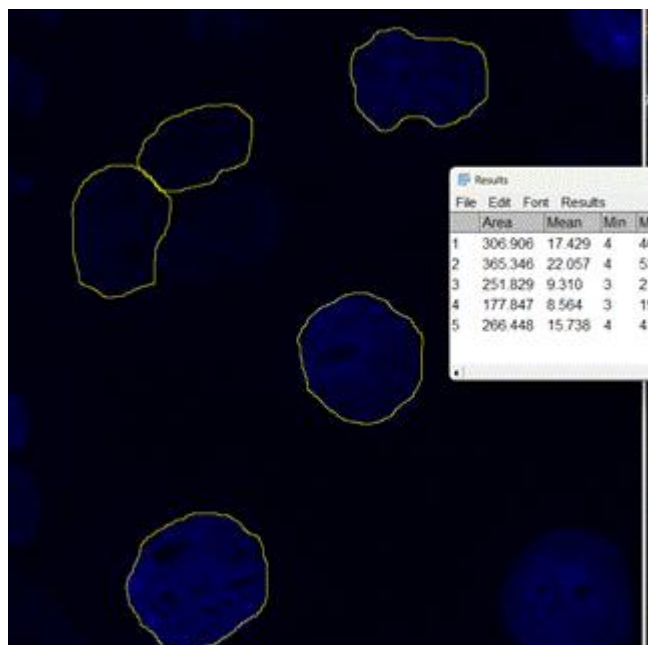


Table 2: Table of the raw measurements and how the area of the nuclei were drawn to make the measurements. Also a couple of attempted cell area measurements, however, these measurements were too uncertain to be properly analyzed.